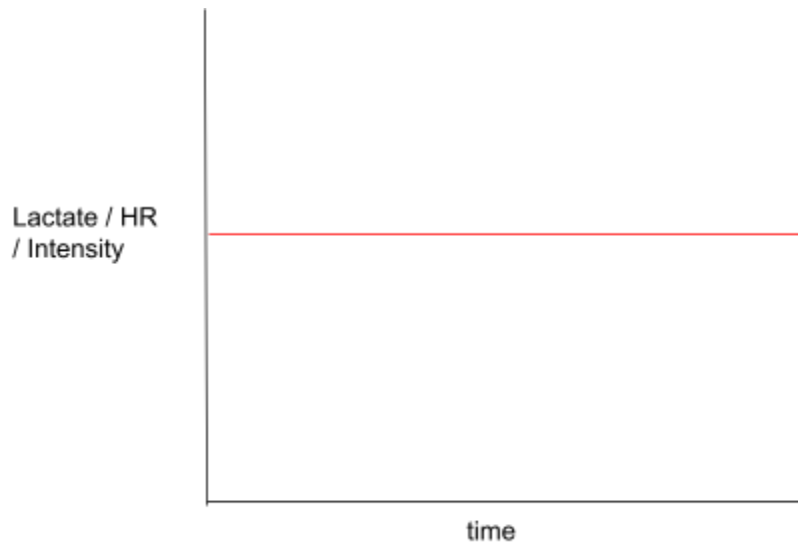
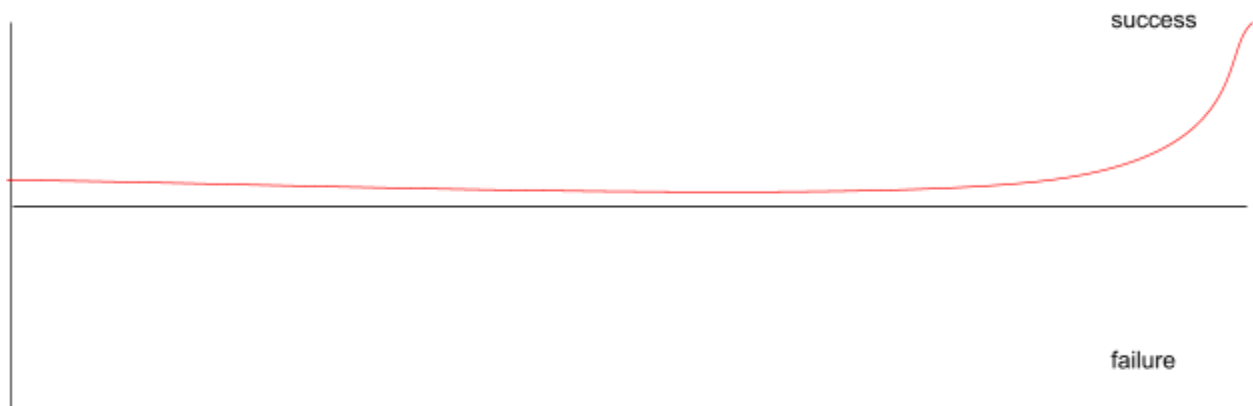


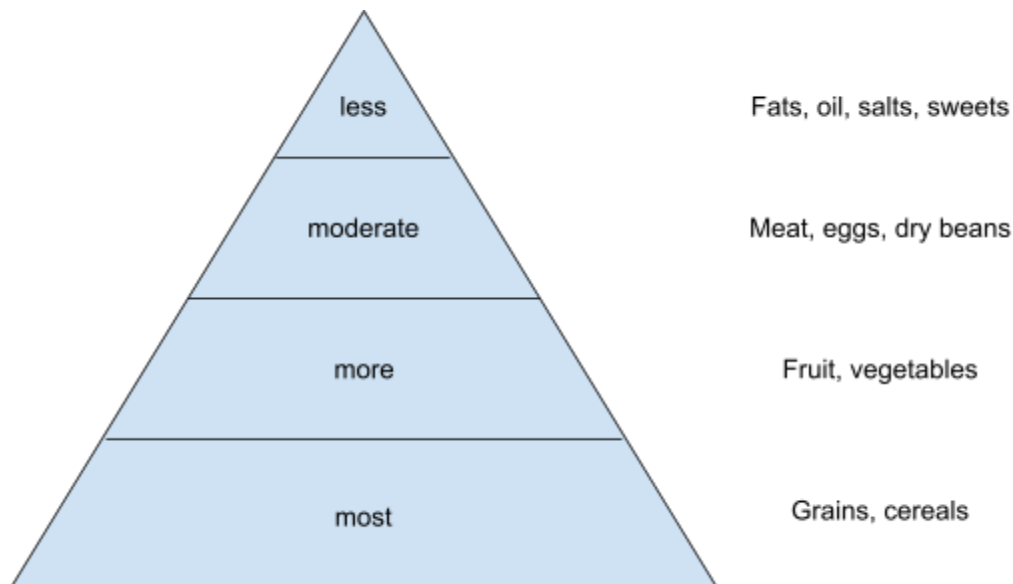
- Peter Snell, John Davies
 - Student-athletes of Lydiard
 - Got coached bc they walked to Lydiard's house
 - The concept behind Lydiard training is that we lack endurance, not speed
1. Aerobic
 - a. O₂ is a catalyst
 2. Anaerobic (not a good system)
 - a. Molecules in muscles w/ O₂ waiting
- Some people are born w/ better anaerobic capacity
 - We need more capillaries → go low intensity/high volume most of the time (Talk Test)
 - Capillary development shifts at 20min, 30min, 55min, 90min, 120min, 150min, 180min
 - Build capillaries and then rest & recover
 - Sleep to win!



Idea of invisible results → exponential growth



- The ant and the elephant
 - Believe that you can be successful
 - Conscious mind is the ant
 - Subconscious mind is the elephant
 - Ant is on elephant's back
 - Positive self-affirmations
- Decide how many hours you spend on each thing each day to help determine how much time you can commit towards the sport
- Prior Proper Preparation Prevents Poor Performance
- Alexi Pappas
 - 33% days are challenging, 33% days are average, 33% days are easy
 - Need at least 9 hours 15 min of sleep
 - Take naps often
 - Iron and vitamin D daily
 - Imitate who you look up to until you can look up to yourself
 - Be kind to yourself just as much as you are hard on yourself
 - Be mad or be happy for 5 min after something and then move on
 - Goals are a long-term process
 - Walk whenever you want to on a run
- 10 rules
 - Take chances
 - Sleep often
 - Dream big
 - Be positive
 - Be nervous in a good way
 - The journey is what matters
 - During races, smile when it hurts
 - Being alone is good sometimes
 - Ask for help when you need it
 - Believe in belief and believe in yourself
- Failure is part of the journey
- Worry about being your own person
- You will never know EVERYTHING
- Nutrition
 - Your body is a machine
 - Green bananas are better for you
 - Try to take at least 1 sip of water every 20 minutes throughout the day
 - Water need (oz) = body weight in lbs / 2
 - Eat everything in moderation



- If it's man-made, it probably isn't good for you
 - Your body needs fat, sugar, and red meat
 - Fats
 - Unsaturated
 - Liquid at room temp
 - Better for you
 - Saturated
 - Solid at room temp
 - Worse for you
 - Carbohydrates
 - Simple
 - Fast energy
 - worse
 - Complex (rice, potatoes, pasta)
 - Long-term energy
 - Better for you
 - Have papaya & pineapple if your muscles are sore
-
- Goal setting
 - Write down a cloud as a goal (not 100% in your control)
 - Write steps as things you ARE in control of to help you reach your goal
 - Positive affirmations are NOT un-negative statements (i.e. I will not procrastinate)
 - They should be positive, present tense, action words, and specific
 - Good example → I always do things on time
 - 10% written down
 - 45% feelings and emotions
 - 45% visualization
 - Share your goals with people who will LIFT YOU UP

- Mantras
 - All senses → bring in feelings and emotions
 - “Light as a feather strong as a _____”
 - “I love hills”
 - “Relaxed yet strong”
 - “Find that strength to finish. There’s no turning back”
 - “I run for a purpose. For life and for strength”
 - “Pity those unprepared for battle for they will never be champions”
 - “Don’t let them take you alive”
 - “Let the fight begin”
- Slight edge jeff olsen
- [Bill Rodgers running log](#)
- What to say when you talk to yourself
- Elite minds
- Resolve orin woodworth
- [NAU Freshman Training Video](#)

Inky Johnson

1. Be happy in the moment & enjoy the process
 2. Learn from defeats & vividly remember victories
 3. You can’t control unexpected objects → you can only control your reaction
 4. Affirmations & goals → see them every day
 5. Visualize every day
 - a. Write. Read. Visualize.
- Always seek new challenges in life
 - If you’re in a big race, you BELONG there
 - Don’t beat yourself up
 - Come up with 3 things you did well for every 1 thing you need to work on

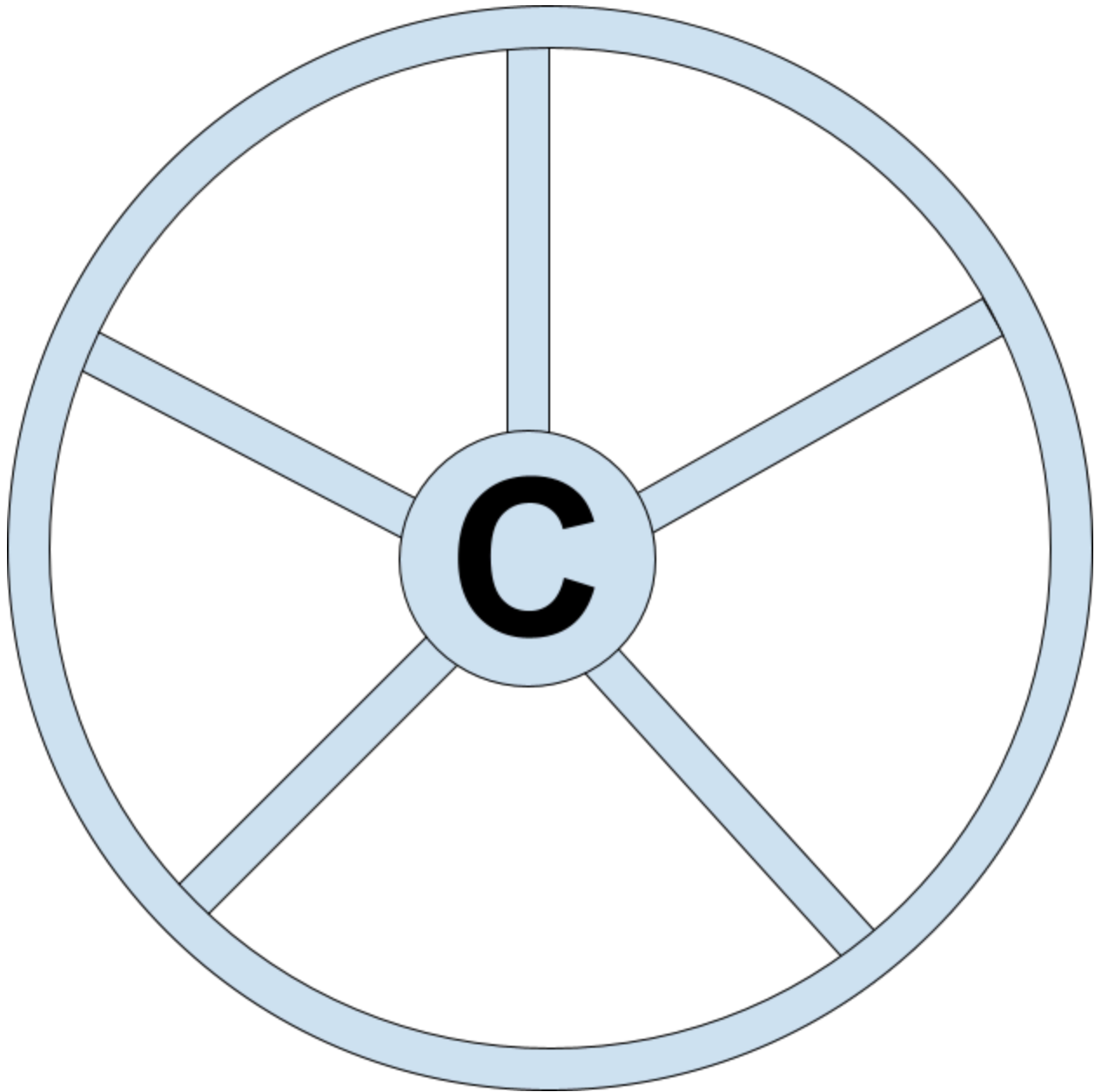
Mental performance plan

- My team at its best:
- Myself at my best:
- To fear and pain:
- Start:
- First third:
- Second Third:
- Final Third:
- Finish:
- What Matters:

Brody Marathon performance plan

- My team at its best: persevere, consistent, healthy
- Myself at my best: Prepared, hard-working, excited
- To fear and pain:
 - Pain
 - Good team-bonding
 - Make it fun
 - Fear
 - Positive self talk (ready to do this)
 - Teammates believe in you
 - Think about it piece by piece
 - Fear can turn into excitement
- Start:
- First third:
- Second Third:
- Final Third:
- Finish:
- What Matters: The journey with teammates

“Wherever we’re going, we’re going together”



“The Wheel” - Enjoy the Grind

- Spokes make up the wheel:
 - Self-investment
 - Goal Setting
 - Prioritize
 - Positive Self Talk
 - Visualize Success
- The wheel once completed is Champions (The C)
- Kick @ 300m left for a shorter race (1K for a longer race)

MEMBERS OF A XC TEAM ARE MUCH LIKE THE SPOKES OF A WHEEL. EACH SPOKE IS NOT ORDERED IN # OF IMPORTANCE, NEITHER IS EACH MAN. EACH SPOKE MUST DO

IT'S JOB OR THE WHEEL WILL NOT REMAIN INTACT. ALTHOUGH ONE SPOKE WILL REACH THE DESTINATION FIRST, WITHOUT THE OTHER SPOKES, THE WHEEL WOULD NOT ARRIVE. YOUR CONTRIBUTION TO THIS XC TEAM IS TO RUN THE BEST RACE YOU CAN. THERE IS NO TIME TO FEEL SORRY FOR YOURSELF IF YOU ARE HAVING A BAD RACE. IF YOU SLOW DOWN OR EVEN WAIT TO MAKE YOUR MOVE. YOU ARE NOT ONLY HURTING YOURSELF BUT HURTING THE TEAM. WE'RE A TEAM, KEEP THIS WORD IN MIND - ALWAYS PUT THE TEAM IN FRONT OF EVERYTHING ELSE.

Mental preparation - a daily practice

- Mentally preparing yourself for daily practice is important and should not be overlooked. The best runners are not always the ones in top shape. Mental attitude separates the champions from the athletes. The following guidelines are to be practiced for daily workouts.
 - Mentally set aside any physical limitations that may hamper the workout
 - Mentally set aside any school work that may occupy your thinking
 - Enjoy each practice session no matter what the pace, course, weather or distance
 - Set mental goals to measure daily practices
 - Support each other (As Olympian Dave Wottle said of his Bowling Green team: "We all train in a group and we draw strength from the group")
 - NOTE: Each day's training will bring improvement. Expect progress to occur on a regular basis. Mentally prepare yourself to excel in training.

Mental attitude during the race

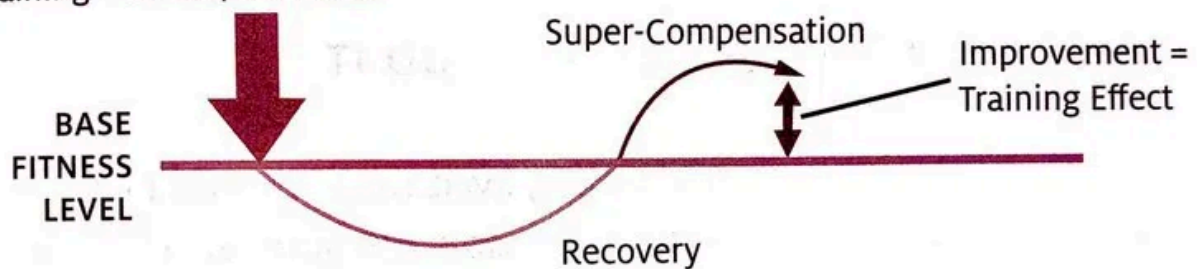
- The following are mental tactics to be used DURING the race:
 - Don't lose concentration. A breakdown in concentration could cost you a place, time or victory. This really means you cease to race and are merely running. Be aware of corners, hills, flats and the finish. Concentrate!
 - Continue to think: " STRONG YET RELAXED."
 - Keep alert to opponents' whereabouts and actions. Watch for tactics to break away from you, pass or lose you at any point. These tactics are designed to mentally upset you. Don't be surprised, but be prepared to counter opponents' tactics.
 - Keep your preconceived race plan uppermost in your thinking, but flexible enough to change if necessary. If you're leading, maintain control: if you are running with a come-from-behind plan, keep alert and know exactly what to do and when.
 - Don't get too excited too early, but don't wait for the last 100 meters to try and get a place → it may be too late

Essential Lydiard

THE ADAPTATION CURVE

- When you subject your body to a training workout, your fitness level temporarily decreases
- During the subsequent recovery period your fitness level will rebound beyond the previous fitness level
- This is called Super-Compensation

Training = Stress/Stimulus



There's an old coaches' principle: *training success = moderate stress + adequate rest*. With any moderate or strenuous activity, fatigue eventually occurs and performance declines. That's followed by an adaptation/recovery phase in which super-compensation occurs, establishing a slightly higher platform of performance for the next period of exertion.

THE KEY TO TRAINING IS TO APPLY APPROPRIATE WORKLOAD
AND RECOVERY SO THAT YOU GAIN STEADY CONTINUOUS IMPROVEMENT.

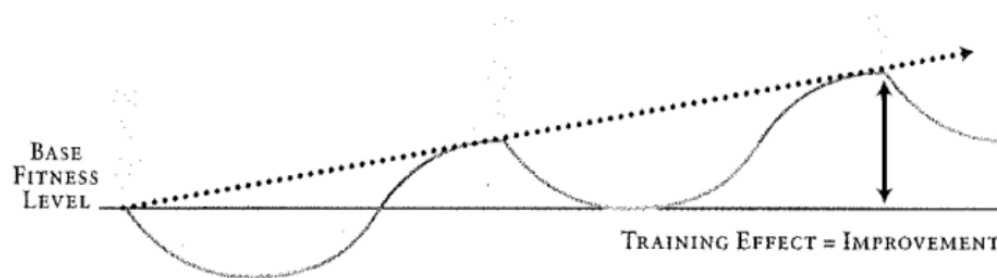


FIGURE 1:
BASIC PRINCIPLE OF ADAPTATION

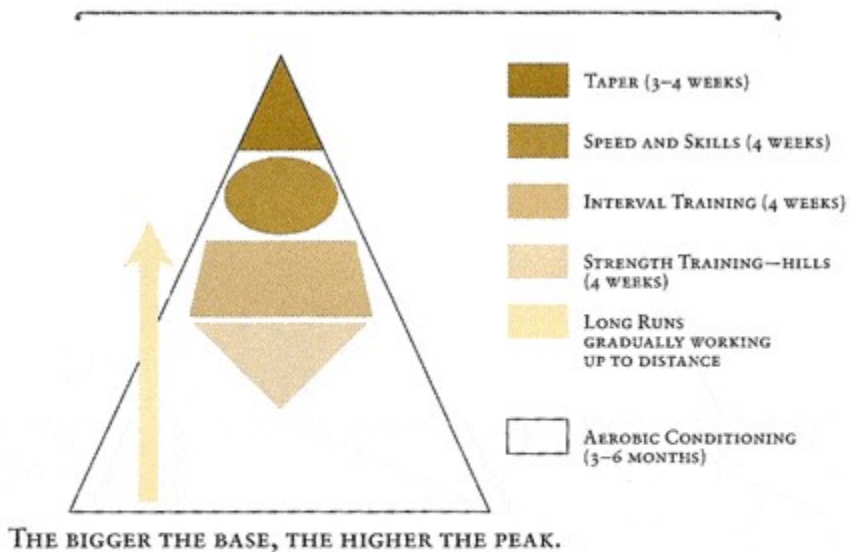
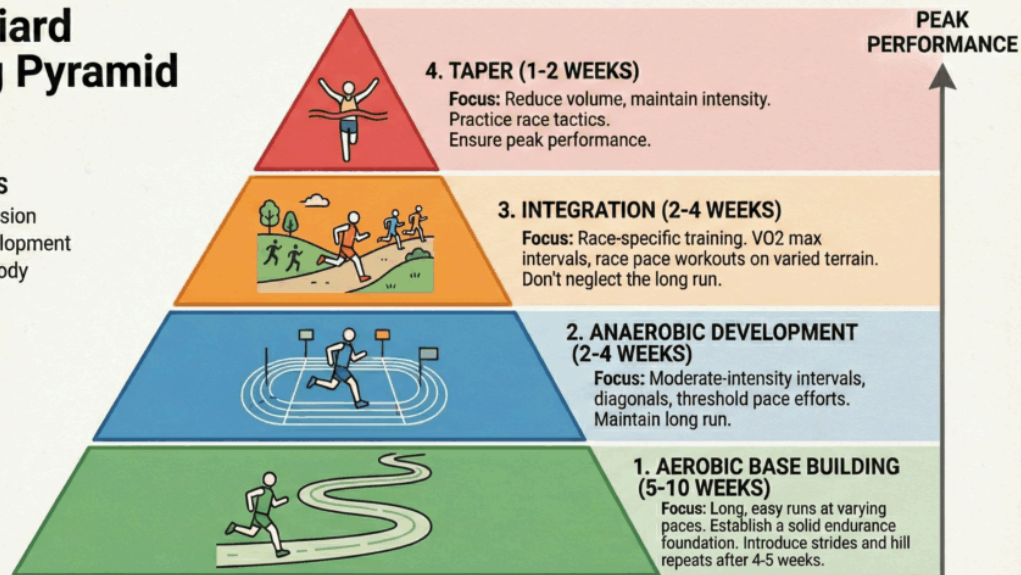


FIGURE 2:
THE LYDIARD TRAINING PYRAMID

The Lydiard Training Pyramid for XC

KEY PRINCIPLES

- Gradual Progression
- Sequential Development
- Listen to Your Body
- Periodization
- Active Recovery



Based on the philosophy of **Coach Arthur Lydiard**: Building a strong aerobic foundation before chasing speed, culminating in race-day readiness.

coachsaltmarsh.com

There's a story told about Arthur Lydiard and his golden boys at the '64 Olympics in Tokyo. Down at the training track, with their rivals looking on, they ran an impressive interval session of 20 quarters. The next day one of these opponents, a talented interval-trained Canadian in his first Olympics, showed up at the track with his coach and proceeded to run the same session. Lydiard's boys cheered him on as he ran each interval faster than the last.

When it was all done a reporter asked Lydiard what he thought of the kid's workout. "I think it was the last nail in his coffin," Lydiard replied.

"But your boys ran the same session yesterday."

"Yes, but my boys needed it."

Perfectly peaked, Lydiard's protege, Peter Snell, won two gold medals. His teammate, John Davies, won a bronze. The Canadian who had eclipsed the Kiwis' training run with his own failed to advance to the finals in his event -- as Lydiard had predicted.

One of my former coaches, Ron Daws, often quoted the above story to illustrate his axiom: "Good training and bad training look exactly the same on paper." Twenty quarters can bring one runner to his or her peak and bury another in a hole. Daws, adhering to Lydiard's philosophy, recognized that stand-alone workouts mean little; it is their contextual application to the advancement of the athlete's goal that matters. Training, then, is not a series of numbers that can be universally applied but is rather the art of combining measure, timing and sequence to the specific needs of the individual. Herein lies the brilliance of the coaching of Arthur Lydiard.

My first coach, John Davies, the same Lydiard protege who won bronze in Tokyo, called Arthur to ask him how to adapt the training for a promising 14-year-old girl. He didn't want to wreck me. Arthur advised him to build a base by increasing my mileage. I began on regular runs, soon reaching my target of 40 miles per week, which I continued for some months. There were never any hard and fast

rules that are regularly attributed to Lydiard, such as the 100-miles-per-week mandate, but rather a careful consideration for my age, gender, ability, fitness and event. And that's how it was for the next 28 years of my career as an international athlete under the guidance of three consecutive Lydiard coaches: Davies, Ron Daws and Dick Quax. My schedules were never replicated in content; however, the principles that defined them as good Lydiard training (and outlined in this article) were unwavering.

1) MILES IN THE BANK: MAXIMIZE AEROBIC CAPACITY FIRST

The first phase of Lydiard training is endurance/aerobic development. Think of aerobic running as home base -- the place where we hang out until we are mature enough to leave, and the place we always come back to for rest and recuperation. As the miles stack up, we increase the capacity of both the heart and the lungs for work, build our circulatory network to the muscles through increased capillarization, increase the number of mitochondria in the muscle cells, and develop other beneficial metabolic and enzymatic pathways for gathering and converting oxygen to energy. Once these structures are established they allow us to respond and recover quickly. Aerobic training, then, is training for all other types of training.

Lydiard often described miles as money in the bank -- the more you have the greater your currency to buy ATP's (the units of energy your muscles need for contraction) and the faster you will eventually be able to race in any event that has a large aerobic component. From the 800 meters on up the body's energy needs are met primarily from aerobic metabolism, with any race over 5,000m more than 85 percent dependent on aerobic ability.

There is no argument: The greatest gain for any aspiring distance athlete is made by spending the majority of time on aerobic development. And for those recreational athletes training for a weekend road race, this is the only phase of Lydiard training that is initially indispensable.

While easy running is always the safest place to start, it is often the case that the cardiovascular system develops quickly while the musculoskeletal system tends to lag somewhat. The golden rule is that you can never progress faster than your slowest part will allow.

Case in point: For the past 20 years Japanese teams have regularly altitude-trained in Colorado. I often recognized some of these petite women, back-packs on, hiking up the steep incline of Gold Hill some 15 miles out of Boulder. Taking the basebuilding strategy one step further, the coach had prescribed extended hiking and jogging sessions during year one of the four-year Olympic cycle solely to condition the musculoskeletal system for base-building. With strong tendons and ligaments these runners could then handle up to 200-mile training weeks without injury, and thus develop a superior aerobic capacity.

2) FEELING-BASED TRAINING: TUNE IN TO YOUR INNER COACH

One of the benefits of the buildup phase is that these longer aerobic runs allow you to safely build a rapport with your body. This is a greatly underrated aspect of training, probably because feelings have generally not been a part of any serious exercise discussion. But being able to precisely gauge one's effort over time is an ability that is the hallmark of all great athletes. They can run the razor's edge, knowing how to pitch their effort and energy to extract the best from their bodies on any given day.

Lydiard fostered this ability in all his runners by prescribing feeling-based training goals. A typical running schedule would ask for half effort on one day and three-quarters or seven-eighths effort on another. Half-effort was half the effort of a full effort, and so on. In the days before heart rate monitors and chronographs one simply had to guesstimate until they got the gist of it, which they did quickly.

Modern technology can be helpful training wheels to developing this rapport but an over-reliance on outside feedback, whether a beeping monitor or a coach bellowing splits from the sidelines, is also a dangerous trap. Ultimately we are on our own in competition, reliant on the clarity of communication between our mind and our body. I call this the "inner coach," the voice within that knows exactly what we need to do at any point in time to reach our potential. Whether it tells us to back off, relax, pick it up, or make a break, in hindsight it was always the right thing to do. As we learn to trust the inner coach over time, the stronger its voice becomes.

3) RESPONSE-REGULATED RECOVERY: BALANCING WORKOUTS

Training can be defined as specific stress applied to the body to invoke a corresponding adaptation. The training stimulus (workout) causes a temporary breakdown in the body (catabolic phase) followed by the adaptive period (recovery), during which the body rebuilds itself so as to better withstand the stress that it has just endured (anabolic phase). Breakdown and buildup better than before, breakdown and buildup better than before; this is the rhythm of good training. Interestingly, the desired training effect does not take place during the workout but during the recovery. We improve not while we are training but while we are resting. More often than not, bad training is a mismatch of breakdown and buildup: either the workout is too hard or the recovery is inadequate, or both.

(Rarely among motivated athletes is it the result of under-training.)

The tendency for many runners to overtrain and/or under-recover is underscored by an inflexible training schedule. While Lydiard pushed his runners, he offset the overtraining syndrome by preparing them for optimal recovery with base training, gearing the training to be feeling-based, and adjusting workouts according to the athlete's recovery response. The art of good training calls for an accurate assessment of which side of the adaptation curve the runner is on -- catabolic or anabolic -- and prescribing appropriately: a recovery run or a workout.

There are simple ways of assessing this: An elevated morning heart rate, poor sleep, low energy, sore muscles and bad mood are all indicators that the runner needs further recovery and a workout of any intensity is contraindicated. Once the "spark" has returned the runner is ready for the next "stress."

Shigeharu Watanabe, coach of Yoko Shibui, whose marathon personal best of 2:19 ranks her No. 7 all-time among women, served his coaching apprenticeship with Lydiard in New Zealand. Of his team he says, "We do only that which is necessary to do, we do not follow any numbers."

4) SEQUENTIAL DEVELOPMENT OF ENERGY SYSTEMS

A study performed by Dr. Izumi Tabata (and colleagues) at the National Institute of Fitness and Sports in Tokyo, Japan, showed that over a six-week period, interval training is more efficient than steady exercise of lesser intensity. This study has been quoted by get-fit-quick advocates as proof of the superiority of interval training. What they do not tell you is that anaerobic development takes four to six weeks to top out, and that when preceded by endurance and strength training the possibility of injury is decreased and the quality of the intervals is enhanced.

Whenever Lydiard was asked which part of the training was the most valuable, he always answered simply, "Everything." Often he would follow up with the question, "Who would want to eat a cake half-baked?" This holistic approach distinguishes Lydiard training from one-brand-fits-all formulas and squelches the quest for the "magic workout." There is no single element that makes one a

champion. Rather, it is the progressive development of each energy system from low intensity to high intensity, each stage preparing the runner for what is to come next. Speed is built on interval training which is built on strength which is built on endurance. When the runner finally reaches her target race, she possesses the entire gamut of adaptive responses from jogging to sprinting.

5) CORRECT TIMING

Peaking is a matter of correct timing. It is one thing to maximize the amount of energy at the disposal of an athlete. It is another to channel that energy into the event that matters. To ensure one's best form is achieved on competition day, a Lydiard schedule is always written from the goal backwards, allotting the amount of time needed for each phase and using the remaining time for base training. Thus the training pyramid serves the twofold purpose of physiological and mental focus to arrive at one point in time -- the goal. There is no need for a separate mental training program of affirmation and visualization -- it is inbuilt into the training program. As the athlete gets closer and closer to the top of the pyramid, the training increasingly simulates the race. There is nothing more confidence-building than the somatic knowing of thorough preparation.

Using the five principles of miles in the bank, feeling-based training, response-regulated recovery, sequential development of the various energy systems and correct timing, the what, when, how and why of every action from sleeping to running is another brick in the pyramid that takes one to the top.

Lydiard's greatest protege, Peter Snell, retired with three Olympic gold medals and numerous world records to his name. In his post-athlete career he came to live in the U.S., where he became a doctor of exercise physiology and associate professor of internal medicine. With the eminent credentials of both athlete and academic, who is better equipped to offer a final thought on the relevance of Lydiard today?

SNELL SAYS:

The core of Lydiard training is the quantity and quality of the base training. Although Arthur appeared to be very precise about the track training leading up to the racing season, I believe this was not as important as the base. Long, moderate-pace running is anabolic whereas high-intensity demanding training, while having its place, is catabolic. Thus the base is critical to prevention of overtraining.

Marathon training for an 800m runner is difficult for many coaches and particularly scientists to understand. Many have been quite dismissive about the benefits in the face of the outstanding results associated with marathon training. To them it makes no sense training slowly for a speed event. The rule of specificity is violated. Why then does it work? Today our knowledge of physiology provides some answers:

-- Long endurance runs appear to provide protection against overtraining from too much high-intensity speed work. Therefore more race-related training may be accomplished.

-- Activation of fast-twitch muscle fibers is normally accomplished by high-intensity interval runs. We now know that long moderate-pace runs also activate fast-twitch muscle fibers, after slow-twitch fibers have become glycogen depleted after the first one to two hours.

In sum, and in light of current physiology, there is little I would do differently today other than incorporate some long, easy intervals once a week during the base training. The training of current top athletes is testimony to the relevance of Lydiard training today.

- **I coach 3 things: the breath, the blood, and the brain**
- **I am still learning and progressing every day as a coach, I change my philosophies slightly all the time**
 - NOT the muscles - most coaches in the US focus mainly on this
 - They coach til your muscles fail - I don't do that - I don't do hard 400s, I don't run til your muscles scream
 - I coach the breath, the blood, and the brain
 - Especially at the high school/collegiate level - the higher you go, the more stuff you will have to add in, but at this level it's the breath, the blood, and the brain
 - *talking to elite high schoolers* "you guys are all still very new to running whether you think you are or not - it takes about 7-10 years to get there"
- The brain portion
 - I am big on my athletes having goals
 - A study of TF olympians had 3 things in common
 - Had a coach
 - Had a team around them to get through the hard days and share the easy days/victories and share the failures
 - Had a training log - at the end of every day you write/type how your training went for that day
 - Can go back and read progress from the past, see how you've progressed
 - ***I make my guys write what your goals for the week are in their journal at the top of every page***
 - I make my guys also write in their journal anything that doesn't go well - whether it be injury niggles, electrolyte problems **AND** write what they think their solutions will be
 - Chart out successes/failures with solving those problems to reference later if they return
 - I make my guys write successes - "today was a good day because...."
 - How did it go and why did it go well
 - **"I am not a big fan of strava"**
 - All you do if you start tracking on strava is starting to **compare yourself with this guy and that guy and so on**
 - STOP doing that - it is YOUR training
 - DO NOT think that "x had a better day on strava than y" - that is a *recipe for disaster*
 - If you have to keep your log on strava, go ahead but know comparison will really build up over time
 - PATIENCE with the log: progress goes up quickly sometimes and then it goes flat
 - Everyone progresses at different rates

- Mental stuff/race strategies
 - I coach to win - I don't care how fast you run I want you to win the race
 - If you win every single race you're in, you'll be very very fast
 - I do not coach to the watch
 - It is a different path for everyone to figure out how to win a race
 - Mentality and race goal is very important
 - I don't let anyone walk to a start line without them knowing what they want to happen at the finish
 - ***I coach the last 25% of a race***
 - When you line up, I don't want you thinking about how you run the first minute or the first half
 - The first race strategy I ever use is trying to figure out how long they can think
 - Lot of brain fatigue generally in the first parts of a race
 - Then when you need the brain in the end, you can't turn it on because you're so mentally fatigued
 - "How long can this kid think? 20 seconds?"
 - You should be able to think for a minute by 11th grade
 - I focus on working on the CLOSE of a race first
 - Once that is figured out, I then "bookend" it - where the brain is on at first, hard open, and then go back to sleep, and then hard finish
 - MUST learn to close first
 - 2024 Olympic Trials
 - Grant running 5/10k
 - Goal was to win the 5k since we assumed he would win the 10k
 - 10k goal: "how could we win and not run hard"
 - In a race, there is always trouble around 50% of the race
 - If 12 people are in the race, you will never hear me say be in 6-10th place by x time
 - Be in either dead last or top 3-5
 - Why? Watch athing mu, shes in 8-10th place and gets tripped and misses out. COACHING ERROR in my mind
 - I don't like guys to lead races early, I wanted grant in top 3-5
 - Strategy: go with 3 laps to go, nobody wound up moving with him from 1000 out
 - The strategy to win that race was to run the last 3 laps hard - he did that and he won without getting too beat up
 - 5,000 prelim: goal is to not win the race but to *play defense*

- 10k was an offensive race: play offense by taking control and winning the race
- Prelim the only thing that matters is top 6
 - **"I don't care what place he is in as long as he is never in 7th"**
- Woody takes off and wins - I don't care if anyone wins a prelim
- Everyone plays defense by making sure they're top 6 and staying out of 7th
- 5000m final: didn't go to plan
 - We pick out our competitors - I know x can't kick with y, I know z is gonna go 800 out, etc
 - Plan was last 1200 in 3:27 - he does exactly that
 - Woody takes off and gets a 50m lead - he didn't have standard so he does this
 - A bunch of people had standards so they weren't gonna play that game
 - Woody doesn't have the capacity to lead an entire race like that
 - I was yelling at everyone else so grant wouldn't have to work in the early part of the race
 - Grant eventually bridges the gap with woody early - bad move
 - **Leading in the middle of the race is rarely one of my race strategies**
 - No matter what, he has to run 3:27 to close no matter how tired
 - Grant had to lead for 3-4 minutes, more wasted than I would like him to
 - He had to run really hard to close like our goal was
 - Our original plan was to lead with 120 to go and really hammer with 80 to go
- **The best thing about Grant is not his speed or his turnover, but his discipline and ability to stay with a race plan**
 - Never "im too tired, I can't keep up"
 - No matter what I know he will try the race plan, he will go at 120 to go if that is the plan
- **When thinking about races: skip the training side of it, think about your brain**
- Building the mental energy to close a race even though you are tired - a skill to work on!
 - Grant's best quality is his brain, he is able to do this effectively
- If you change your race strategy mid race, you are toast!
 - You can beforehand
- "Grant is not fast!" he could barely run 50 today
 - However, when it comes to racing, grant is not tired
 - If i tell him to run a 56 second closing 400 and by the time he gets there he is not tired
- Why? Because he trains his lungs - his breath and blood

- He is super super super strong and his aerobic training is really dialed in so when it comes to the end of the race he can really turn it on
- One more story - not about grant this time
 - I figure out the puzzle and I want to know what pieces there are and then I have to put together the pieces to make someone a fast runner
 - Everyone is a different type of puzzle
 - The best for one isn't the best for another - especially for HS kids
 - I have this 17:37 5k kid
 - In my world 17:37 as a junior is slower
 - I get him in February
 - Hes kinda brain dead so its pretty easy to coach him
 - He knows everything and is stubborn as can be
 - I bet him 50 bucks I could make him top 50 in the state on the condition that he shuts his mouth and stops asking why and saying "but coach"
 - Especially if you're that bad of a runner
 - "The fast guys have already had dinner by the time he crosses the line"
 - We start training and I did one thing with him, only aerobic training with him - a tempo run monday wednesday and friday with a weekend long run
 - EVERY WEEK
 - This is all we did and he ran 15:43 in cross and got on the team at michigan state
 - I have another kid
 - Your bodies are probably better trained than your minds are
 - Jude - one of his freshman
 - "Jude your goal freshman year is to break 4:40"
 - "Coach I have never even broken 5"
 - Im talking to a kid who has run 5:05 and now I am telling him he is gonna run 4:39 - it's totally unreasonable
 - He starts telling himself that he can run 4:39
 - First time out he runs 4:46
 - He finished the freshman season at 4:36
 - Jude's brain was at 5:05 and his brain finally met up with his body!
 - I had a sophomore and I set him with a race plan to PR in the mile
 - Unfortunately for him it was the first mile of a 2 mile
 - He wound up PR'ing the 2 mile and placed 3rd in the state as a senior
-
- Questions and Answers about training
- *What do you do with an athlete who is a little more on the OCD side, like run as much as possible and does all the extra stuff, how do you coach their brain?*

- OCD tendencies whether or not anyone wants to admit it is very prevalent in distance runners. Being perfect is a dangerous thing to do. The body is way smarter than me, and I am a good coach. With OCD tendencies and being perfect, I like to make fun of failure and reduce the internal pressure that one is feeling. If you're setting yourself up to be perfect in training - you're done. No training is perfect and no day is perfect. But be really really good and you've got to be in a position where you can laugh at yourself and walk away and say "that was really good and it didn't have to be perfect. Pressure is an internal thing. Someone can tell me that I have to win this race - whether or not I internalize that is my decision. You have your own brain, you use it yourself. You line up with a race strategy. Going back to grant with the 5k at the trials. I told him to move at 80 and it didn't happen. It wasn't perfect, but it was really good and he still won. Give yourself a little bit of a break and relax that. I relax the expectation at races - for me if that was going on, I would change it so that the brain is thinking of different portions of the race. I don't do perfection with any goals - athletic, grades, you name it. I will say run 6 miles easy and then we'll get back at 5.81 miles and because everyone's garmins are different everyone will run various miles until they see a perfect 6 and I want to break the things. 6 miles easy is just a stimulus to prepare for the harder session, so you know if you're at 5.8 - its the same thing. It doesn't have to be perfect. I am not a big strava or garmin guy, you know this.
- *How could you tell Jude was ready to run fast?*
 - I watch every step of every workout. As I am watching, I am seeing his body and mechanics work, how efficiently can he go across the ground. Also, how much of his mind is at work. Is he able to sleep through it and still go across the ground? Is he relaxed at certain things. With him, 4:39 was the goal and I should have had it at 4:29. If I had another month I could have gotten him there. Breaking 5 even is a big mental thing as is breaking 440. 4:39 is a high school freshman is proper training, not spectacular but proper. Overall, I know you are all HS'ers and you're greedy and you want to run PRs every race but I don't coach that way really. I care about the long term development, I want you all sitting in the same chairs grant is sitting in. I don't coach for a season, I coach you for THEN.
- *How do you view nutrition?*
 - The way I coach, every kid is a sponge. I pour a little water on the sponge, and if it gets absorbed, I pour a little more on and so on. Everything you do with a kid nutrition wise, they have to want to do it - whether its nutrition or training. Every kid can improve their diet, they just don't know how to do it. I preach no label diet, which means that you're going to do nutrition right. You can eat whatever you want, as much as you want, as long as it doesn't have a label on it. Go find something without a label on it - you'll find them in the fruits and veggie aisle as well as the meat aisle. I prefer lower to the ground food and stuff with no labels on it. As far as carbohydrates,

proteins, and fats - you need to give them information because high schools are really stupid. You need to teach them that carbs digest faster than proteins or fat, so you would want to eat them before training or racing. There is an elimination process with eating, and that better be 3, 4, 5 hours or something like that and when you eat before racing that better be around 3-400 calories and a carbohydrate so that you have little in the stomach when the gun goes off as the stomach prioritizes digestion over running fast. You don't want any of your energy going to digestion. Experiment and put your findings in your log.

- *What about nutrition for early races?*
 - So then you have to reduce it down to like 1-200 calories and you can drink those calories as it gets them in quicker than eating. Most of my guys do the drink mix before an 8 am or 9 am race rather than eating. Pros in general will jog a mile 6 hours before a race. They'll come back and shower then come back and eat 2-300 calories. Grant eats oatmeal, some eat toast. Very little simple sugars - really just that. Like jam, anything that's actually a sugar when you put it on something. This is usually eaten 4 hours before a race. At the highest level, there is a sequester period where you have to go in a tent 30 mins before. 75 mins before a race we warmup and then they go into the tent.
- *What are ways to work on confidence during races?*
 - The number one thing is to be your best friend. Like I said earlier, keep a log. In that log I want you to write your successes. Repeated successes in the log = repeated successes on the track. What happens in life today with social media posts and all that crap, everyone tells you you can't do it and you're not good enough. You need to turn the tables on that, throw strava away and you need to create your best friend. That best friend better be looking at you in the mirror. Write in the log that you succeeded today and that you did everything you needed that day. When you run a race, you write what went well. At the end of the season my guys and girls always want to tell me where they failed. I don't want to know where you failed - I want to know what you did right that day. Did you sleep well? Did you eat well? Did you run easy enough today? You make your own life based on the smallest of decisions made each day. Over time and after 1000s of good decisions, you make a beautiful life. Don't ignore the small decisions - because NOT making one is also a decision. I want you to celebrate your good decisions and do it in your log by yourself. Do this every single day - make success a habit. Success isn't everyone else falling off the pace and you winning - that is not the way you win. Success is a habit, create that habit with yourself in that log every single day. And then, you can translate that into races. It is a slow process, however, as races are really hard on the brain. I am ok with negative thoughts coming in your brain by the way. You have to say they're going to appear, you can't say they're never coming in. This is where you come to play, you move that negative energy

out and get back to what you are trying to accomplish in the race and replace it with a positive thought. Practice that starting out today so when the race comes it's easier to move the negative thoughts out.

- *In terms of training and racing, how did you figure out what you liked best in coaching?*
 - **Arrogantly I will say I am a pretty good coach. I believe that I was such a bad and failed runner that I kept trying to figure out how to succeed. I wouldn't quit - I'd try this system, and another, and another. I kept failing. So I am the greatest failure of all of you guys. What I learned was how the body reacts and adapts and I applied my failures to your successes. That is the absolute truth. I could show you my log entries - if you read them you'd think I was going to kill someone. I was really mean to myself and I was really negative but I learned through failure. I don't care if you fail but you've gotta learn from it and get back up on your horse. In my opinion, I did run fairly fast a few times and got to wear the USA uniform, but I learned from failure and from other guys who were faster than me. Honestly, I am still learning to this day. My guys who are on the same team don't have the same training plans so I learn from that quite a bit. You can't stop learning. If I was faster, I'd be a worse coach.**
- *What do you think about cross training and Parker Valby?*
 - **I don't know why she is so popular. I will tell you that Parker, in my opinion, might train a little more than you think she does. She is making waves by promoting that she doesn't really train. As far as cross training goes, it depends how far you want to go. If you have a bone density issue, I am a big fan of cross training. However, if you're going to cross train, you have to mirror what the actual training would be on that day. If you're cross training on an easy day, just go do an easy cross train. However, if you're missing a tempo or harder effort, you just adopt the same minutes at the same effort. The hardest thing to do is swimming, because you're horizontal so the HR is different and the rest is 6:1, so if you swim hard for a minute you get 10 seconds. You can't do that in running. The bottom line is whatever you're going to do that day running, you do the same cross training. For the most part, you need to strike the ground. If you can't strike the ground, you're going to need cross training. If you're gonna be a runner, be a runner first and foremost.**
- *Can you outline what the most efficient tempo run is?*
 - **Boy that is tough. I have all my high school guys run 20-24 minutes of tempo running in a workout. You start blowing up somewhere between 7-9 minutes in, so we always are adjusting based on a number of conditions. I ask my guys to rate out of 10. When they start running harder than a 7/10, things get hairy. When they want to complain "coach, this is way too easy!" I know they're in the right place. That is pretty much how a tempo run works honestly. If you complain that you're running too hard, I might let you run one more interval, but I won't let you cross that 7/10 threshold. I am guessing everyone runs their tempos too fast. 6 out of 10 to me is a proper**

tempo run, 20-24 minutes of duration of that. 4 or 5 x 5 minutes is perfect. Even for VO2 stuff, all of our stuff is 24 minutes max. If you go too hard and only get 12 minutes in, and your buddy does it right and gets 24 minutes in, well he is going to kick your butt in about 6 months.